

Assignment 2

- i) Find out and discuss how a linear regression model uses categorical variables in model creation?

Response:

First of all, the linear regression model takes the first level of each factor (categorical variable) as the reference category. For "cut", the "Fair" is the reference. For "colour", the "I" is the reference. For "clarity", the "I1" is the reference. By default, the "cut" and the "clarity" have their worst quality as the first level in the factor. To be consistent with that, I re-ordered the level of the "colour". Therefore, all the coefficients of all categories are positive.

We assume the coefficient of the reference category is 0. Take the "cut" as an example, if the other variables are staying the same, the price will increase by 611.84 with the "cut" changing from "Fair" (reference category) to "Good".

```
> summary(mymodel)
```

Call:

```
lm(formula = price ~ carat + cut + colour + clarity, data = dia)
```

Residuals:

Min	1Q	Median	3Q	Max
-8894.8	-678.1	-191.9	454.4	7071.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9355.76	185.33	-50.482	< 2e-16 ***
carat	8932.20	38.70	230.831	< 2e-16 ***
cutGood	611.84	108.58	5.635	1.85e-08 ***
cutIdeal	926.02	99.27	9.329	< 2e-16 ***
cutPremium	786.05	100.05	7.856	4.81e-15 ***
cutVery Good	826.51	101.07	8.178	3.63e-16 ***
colourI	1046.30	92.36	11.329	< 2e-16 ***
colourH	1452.28	88.18	16.469	< 2e-16 ***
colourG	1945.22	86.74	22.427	< 2e-16 ***
colourF	2079.75	87.92	23.654	< 2e-16 ***
colourE	2241.71	88.52	25.323	< 2e-16 ***
colourD	2387.33	92.49	25.812	< 2e-16 ***
clarityIF	4767.44	167.01	28.546	< 2e-16 ***
claritySI1	3167.56	143.91	22.011	< 2e-16 ***
claritySI2	2218.76	144.89	15.313	< 2e-16 ***
clarityVS1	4155.91	146.83	28.305	< 2e-16 ***
clarityVS2	3811.46	144.82	26.319	< 2e-16 ***
clarityVVS1	4667.13	154.68	30.172	< 2e-16 ***
clarityVVS2	4430.33	151.27	29.288	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1133 on 4981 degrees of freedom

Multiple R-squared: 0.9209, Adjusted R-squared: 0.9206

F-statistic: 3221 on 18 and 4981 DF, p-value: < 2.2e-16

```
> |
```

- ii) Except for carat, which ONE of the other attributes (i.e. clarity, colour or cut) has the greatest effect on price? Which one has the least effect on price?

Response:

To answer this question, I would apply the method of calculating the mean of the differences between adjacent coefficients of each categorical variable. More specifically I apply the built-in functions of *mean* and *diff* in R.

For the cut, the mean of differences of coefficients is 231.505.

```
> mean(diff(c(0,611.84,786.05,826.51,926.02)))  
[1] 231.505
```

For the colour, the mean of differences of coefficients is 341.0471.

```
> mean(diff(c(0,1046.30,1452.28,1945.22,2079.75,2241.71,2387.33)))  
[1] 341.0471
```

For the clarity, the mean of differences of coefficients is 681.0629.

```
> mean(diff(c(0,2218.76,3167.56,3811.46,4155.91,4430.33,4667.13,4767.44)))  
[1] 681.0629
```

On average, each leveling up of clarity causes 681.0629 of price increase with the same carat, cut and colour, followed by 341.0471 for the colour and 231.505 for the cut.

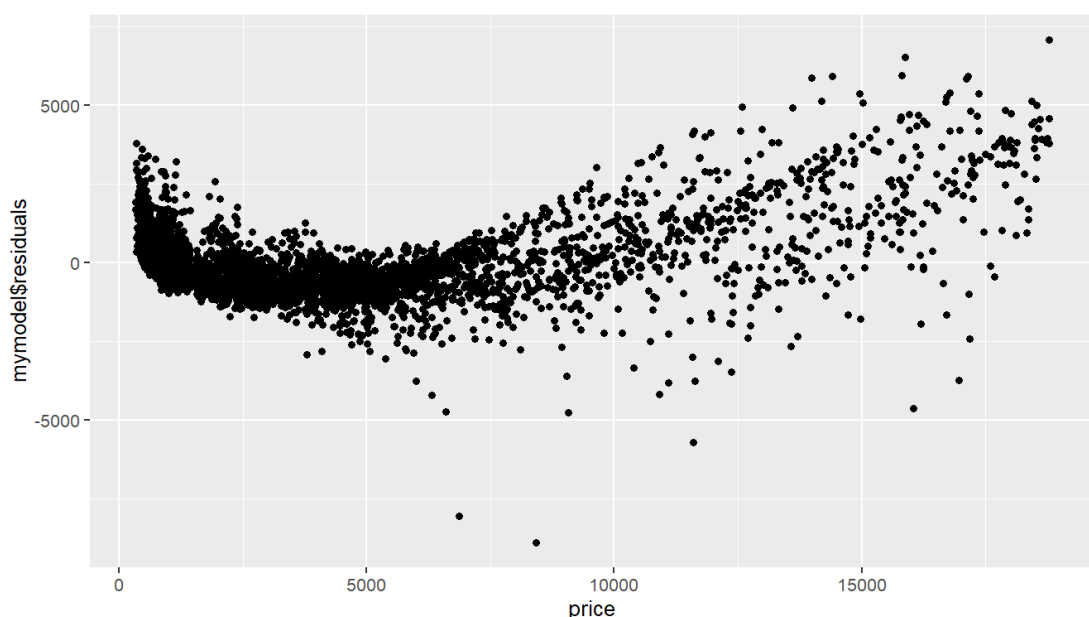
Therefore, in general the clarity has the greatest effect on price and the cut has the least effect on price.

- iii) Perform a residual analysis of the model and evaluate whether the linear model is an appropriate and valid model of diamond prices.

Response:

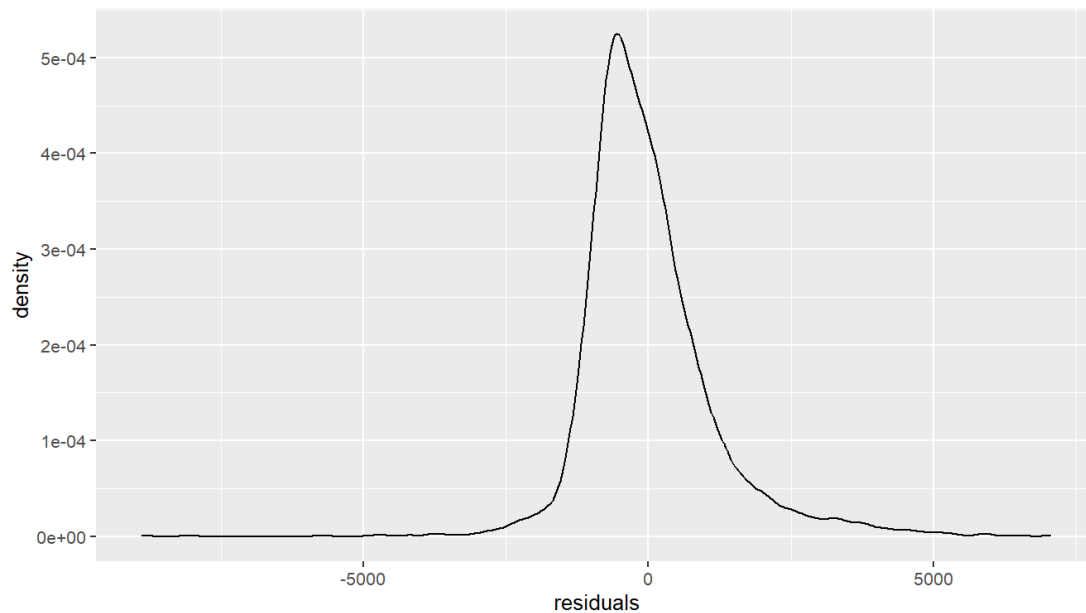
- 1) Scatter-plot of residuals:

Scatter-plot shows a clear curved pattern rather than a random distribution. It means that the linear regression model does not perfectly explain the relationship between predictors and diamond price.

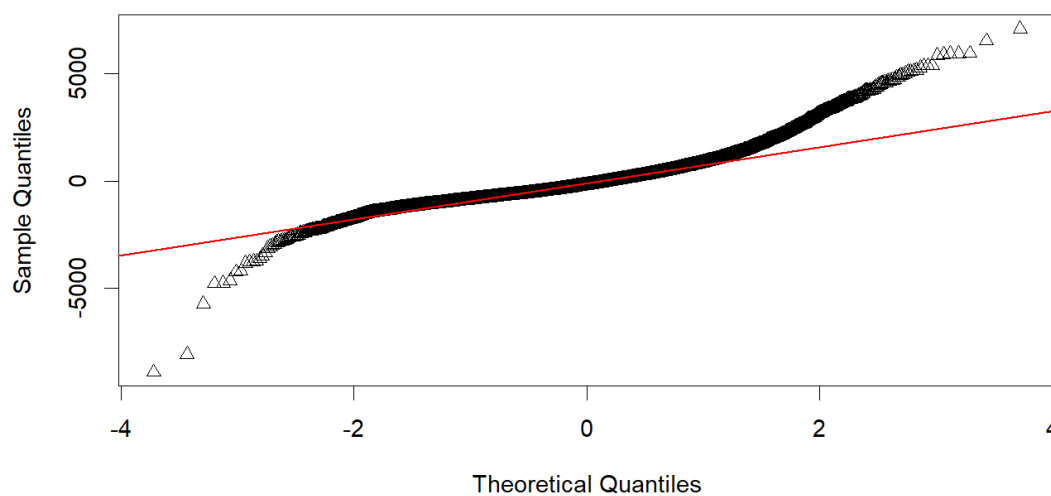


- 2) Density plot and Normal Q-Q plot:

The density plot shows that the residuals are not normally distributed. The Normal Q-Q plot shows substantial deviations from the two tails of the reference line, suggesting that residuals do not follow a normal distribution.



Normal Q-Q Plot



3) The Shapiro test:

The Shapiro test indicates by W and p-value that the residuals are not normally distributed.

```
> shapiro.test(mymodel$residuals)
```

shapiro-wilk normality test

```
data: mymodel$residuals  
W = 0.90098, p-value < 2.2e-16
```

In conclusion, the linear model is not an appropriate and valid model of diamond prices.